

1604

## Irrigation and Multipurpose Projects of India

About 1 question a paper from the Indian Geography slot, but the same material comes back a second time through the Rajasthan Geography slot - IGNP and the Chambal projects are asked almost every year there. Atal Bhujal Yojana alone was asked in 2021 and again in 2023.

India's monsoon works for four months and then stops, so almost every serious crop in this country stands on irrigation. RPSC does not ask you to analyse that. It asks project-to-river, project-to-state, scheme-to-year. Learn this chapter as a set of pairs and you will not lose a mark here.

### Irrigation and Multipurpose Projects

#### Sources

- Tube wells ~45%
- Canals ~25%
- Other wells ~16%
- Tanks ~3%

#### Joint projects

- DVC 1948
- Kosi
- Gandak
- Chambal
- Bhakra-Nangal

#### Water harvesting

- Khadin
- Johad
- Zabo
- Ahar-pyne
- Kuhl

#### North Indian dams

- Bhakra-Sutlej
- Tehri-Bhagirathi
- Rihand-Son basin
- Pong-Beas

#### Rajasthan link

- Indira Gandhi Canal
- Harike barrage
- Sardar Sarovar share
- Chambal valley

#### Interlinking

- NPP 1980
- NWDA 1982
- Ken-Betwa

#### Peninsular dams

- Hirakud-Mahanadi
- Nagarjuna Sagar-Krishna
- Tungabhadra
- Ukai-Tapi

#### Schemes

- PMKSY 2015
- CADP 1974-75
- Atal Bhujal 2019
- Jal Jeevan 2019

Where India's irrigation water actually comes from

Before the projects, fix the arithmetic of sources. Net irrigated area is the land watered at least once in a year. Of that area, tube wells and other wells together do the heavy lifting - more than 60 per cent. Canals come second. Tanks, which every textbook talks about, are today a very small share. RPSC's favourite trick in this chapter is to inflate the tank figure and see if you notice.

### Sources of irrigation by share of net irrigated area

| Source        | Approximate share | Where it dominates                             |
|---------------|-------------------|------------------------------------------------|
| Tube wells    | About 45%         | Punjab, Haryana, western UP, Bihar             |
| Canals        | About 25%         | Northern plains, deltas of the east coast      |
| Other wells   | About 16%         | Peninsular hard-rock areas                     |
| Other sources | About 11%         | Scattered, includes lift and diversion         |
| Tanks         | About 3%          | Peninsular plateau - Tamil Nadu, Telangana, AP |

- Uttar Pradesh has the largest net irrigated area of any state.
- Punjab has the highest share of its net sown area irrigated - about 98 per cent.
- Rice takes the largest share of India's irrigated area; wheat is second.
- Wells plus tube wells together water over 60 per cent of the irrigated land.

### Canal country vs tank country

| Canal irrigation                                        | Tank irrigation                                      |
|---------------------------------------------------------|------------------------------------------------------|
| Northern plains - flat land, perennial Himalayan rivers | Peninsular plateau - uneven rock, seasonal rivers    |
| Alluvial soil, easy to dig, water holds level           | Hard crystalline rock, natural hollows used as ponds |
| Punjab, Haryana, UP, Bihar, coastal deltas              | Tamil Nadu, Telangana, Andhra Pradesh, Karnataka     |
| About one-fourth of net irrigated area                  | About 3 per cent and falling                         |

**TRAP**

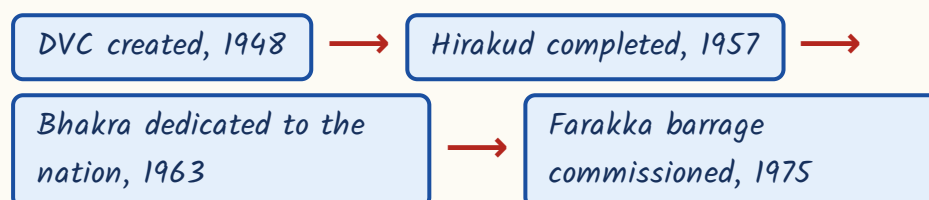
Tanks irrigate about 3 per cent - not one-fourth. That inflated figure is a standard wrong statement in statement-based questions. Canals are the one-fourth.

## What 'multipurpose' means, and the project that started it

A multipurpose project is one dam doing several jobs at once - irrigation, hydro power, flood control, navigation, fisheries, drinking water, sometimes tourism. Nehru called these dams the temples of modern India. The first one built after independence was the Damodar Valley Corporation, and it was consciously copied from America's Tennessee Valley Authority. Remember DVC as the debut.

- DVC set up in 1948 - first multipurpose river valley project of independent India.
- Model: Tennessee Valley Authority of the United States.
- Dams on the Barakar - Tilaiya and Maithon; on the Damodar - Konar and Panchet.
- It also runs the Durgapur barrage; the states served are Jharkhand and West Bengal.
- The Damodar was called the 'Sorrow of Bengal' before the project tamed its floods.

## The order in which the big four came up - a favourite ordering question



## The big dams - river, state and the one fact that identifies each

This is the core table of the chapter. RPSC asks it as project-to-river or project-to-state matching, and sometimes as a negative question - 'which pair is NOT correctly matched'. Do not try to remember heights for all of them. Remember one identifying label per dam, because that is what the option strings carry.

### Project - river - state (set 1)

| Project                      | River                     | State(s)                    |
|------------------------------|---------------------------|-----------------------------|
| Bhakra                       | Sutlej                    | Himachal Pradesh (Bilaspur) |
| Nangal                       | Sutlej, below Bhakra      | Punjab                      |
| Tehri                        | Bhagirathi and Bhilangana | Uttarakhand                 |
| Rihand                       | Rihand, a Son tributary   | Uttar Pradesh (Sonbhadra)   |
| Pong (Maharana Pratap Sagar) | Beas                      | Himachal Pradesh            |
| Ranjit Sagar (Thein)         | Ravi                      | Punjab-J&K border           |
| Nathpa Jhakri                | Satluj                    | Himachal Pradesh            |
| Farakka barrage              | Ganga                     | West Bengal                 |
| Hirakud                      | Mahanadi                  | Odisha                      |
| Sardar Sarovar               | Narmada                   | Gujarat                     |

### Project - river - state (set 2)

| Project           | River       | State(s)                     |
|-------------------|-------------|------------------------------|
| Nagarjuna Sagar   | Krishna     | Telangana / Andhra Pradesh   |
| Tungabhadra       | Tungabhadra | Karnataka and Andhra Pradesh |
| Sriramsagar       | Godavari    | Telangana                    |
| Mettur (Stanley)  | Kaveri      | Tamil Nadu                   |
| Idukki (arch dam) | Periyar     | Kerala                       |

| Project                                        | River   | State(s)         |
|------------------------------------------------|---------|------------------|
| Koyna                                          | Koyna   | Maharashtra      |
| Ukai                                           | Tapi    | Gujarat          |
| Bansagar                                       | Son     | MP, UP and Bihar |
| Gandhi Sagar                                   | Chambal | Madhya Pradesh   |
| Rana Pratap Sagar, Jawahar Sagar, Kota barrage | Chambal | Rajasthan        |

- Tehri - 260.5 m, the tallest dam in India.
- Bhakra - 226 m, the highest straight gravity dam in India; reservoir Gobind Sagar.
- Hirakud - with its dykes about 25.8 km, among the longest dams in the world; built 1957.
- Nagarjuna Sagar - one of the largest masonry dams; canals Jawahar (right bank) and Lal Bahadur (left bank).
- Rihand - reservoir Govind Ballabh Pant Sagar, one of India's largest artificial lakes.
- Farakka, 1975 - diverts Ganga water into the Bhagirathi-Hooghly to flush silt and keep Kolkata port navigable.
- Sardar Sarovar - the Narmada main canal is among the world's largest irrigation canals.

#### TRAP

Two Bhakra traps in one box. First, Gobind Sagar is Bhakra's reservoir on the Sutlej, while Govind Ballabh Pant Sagar is Rihand's in Uttar Pradesh - the names look almost identical in an options list. Second, Bhakra is on the Sutlej, never on the Beas; 'Bhakra - Beas' is the standard planted wrong pair. The Beas dam is Pong.

## Joint and inter-state projects

Rivers do not respect state lines, so several projects are run jointly. RPSC likes these because the answer is a list of states, which is easy to scramble in an option. Two of them cross an international border - Kosi and Gandak both involve Nepal, the Kosi being the river called the 'Sorrow of Bihar'. Note also that Rajasthan appears twice here, as a partner in Bhakra-Nangal and as a beneficiary of Sardar Sarovar.

### Project to participating states

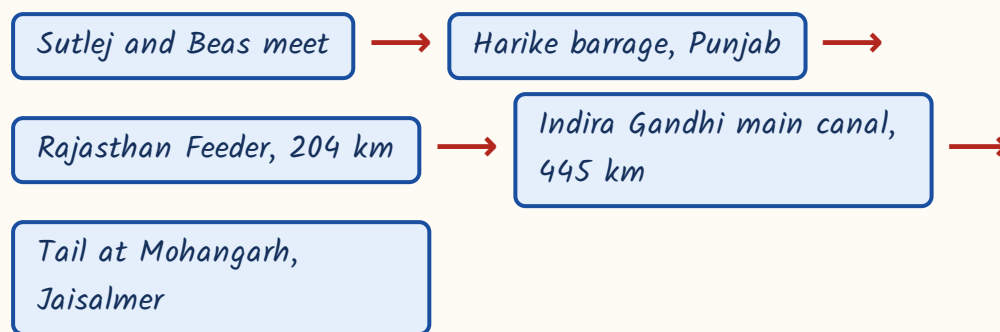
| Project             | Participating states / countries                |
|---------------------|-------------------------------------------------|
| Bhakra-Nangal       | Punjab, Haryana, Rajasthan, Himachal Pradesh    |
| Damodar Valley      | Jharkhand and West Bengal                       |
| Kosi                | Bihar and Nepal                                 |
| Gandak              | Bihar, Uttar Pradesh and Nepal                  |
| Chambal valley      | Madhya Pradesh and Rajasthan                    |
| Tungabhadra         | Karnataka and Andhra Pradesh                    |
| Parambikulam-Aliyar | Tamil Nadu and Kerala                           |
| Bansagar            | Madhya Pradesh, Uttar Pradesh and Bihar         |
| Sardar Sarovar      | Gujarat, Madhya Pradesh, Maharashtra, Rajasthan |

- The Chambal valley project has four structures - Gandhi Sagar, Rana Pratap Sagar, Jawahar Sagar and the Kota barrage.
- Nangal Hydel Channel feeds the Bhakra main line canal.
- Kosi is Bihar plus Nepal; Gandak adds Uttar Pradesh to the same pair - the shorter name has the shorter list.

## The Indira Gandhi Canal - the one project Rajasthan is asked about

This is the longest canal in India and the single most examined irrigation work in the whole RAS paper. It was built to carry Punjab's rivers into the Thar. It was originally called the Rajasthan Canal and was renamed after Indira Gandhi. Learn the water's route from source to tail - questions are built on exactly that sequence.

### The route of the water



- Off-takes from the Harike barrage, just below the Sutlej-Beas confluence.
- It carries the waters of the Sutlej and the Beas, not the Ganga system.
- Rajasthan Feeder 204 km plus main canal 445 km - the longest canal in India.
- Built in two stages; several lift canals take water uphill from it, including Kanwar Sen (Gharsana), Pannalal Barupal and Chaudhary Kumbha Ram.
- The command covers the north-western districts, from Sriganganagar and Hanumangarh down to Bikaner, Jaisalmer and Barmer.
- Its criticism is waterlogging and rising salinity in parts of the command.

### ASKED IN RAS

Asked in RAS Pre 2023 - the Indira Gandhi Canal takes off from which barrage? Answer: Harike. If you remember nothing else about this canal, remember Harike.

### TRAP

The main canal does not end at Sriganganagar. It runs south-west to Mohangarh in Jaisalmer. Sriganganagar sits on the older Gang Canal system, commissioned in 1927.

## Schemes - name, year, and the one line that identifies each

Scheme questions are pure recall. The examiner picks the launch year, the ministry or the component list. Learn those three things for each scheme and skip the rest of the brochure.

### Irrigation and water schemes

| Scheme                             | Year             | Identifier                                     |
|------------------------------------|------------------|------------------------------------------------|
| Command Area Development Programme | 1974-75          | Now CAD and Water Management, under PMKSY      |
| National Water Policy              | 1987             | Revised in 2002 and again in 2012              |
| National Water Development Agency  | 1982             | Executes the National Perspective Plan of 1980 |
| PMKSY                              | 1 July 2015      | Motto 'Har Khet Ko Pani'                       |
| Atal Bhujal Yojana                 | 25 December 2019 | Groundwater, Ministry of Jal Shakti            |
| Jal Jeevan Mission                 | 2019             | 'Har Ghar Jal' - household tap connections     |

- PMKSY 2021-26 phase outlay: Rs 93,068.56 crore.

- Its three components are AIBP, Har Khet Ko Pani, and Watershed Development.
- 'Per Drop More Crop' - drip and sprinkler - was a PMKSY component till 2021-22.
- From 2022-23 Per Drop More Crop is implemented under the Rashtriya Krishi Vikas Yojana.
- The Micro Irrigation Fund sits with NABARD.
- The National Water Mission under NAPCC targets a 20 per cent improvement in water-use efficiency.

### Atal Bhujal Yojana - the most repeated scheme in this chapter

- Launched 25 December 2019, on Vajpayee's birth anniversary; Ministry of Jal Shakti.
- Outlay Rs 6,000 crore - half a World Bank loan, half the Central share.
- It is demand-side and community-led: gram panchayats prepare water security plans.
- It covers only seven water-stressed states - Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh.

#### ASKED IN RAS

Asked in RAS Pre 2021 - which ministry implements Atal Bhujal Yojana?

Answer: Jal Shakti. Asked again in RAS Pre 2023 as a statement question, where the wrong statement was 'it is being implemented in all the states of India'. Two papers, one scheme.

#### TRAP

Punjab is NOT in the Atal Bhujal list, even though its groundwater is among the worst in India. The seven states are Gujarat, Haryana, Karnataka, MP, Maharashtra, Rajasthan, UP. Mnemonic: GH-KMM-RU.

## Groundwater, traditional harvesting and river interlinking

India draws more groundwater than any country on earth, so the Central Ground Water Board's Dynamic Assessment has become a standard source of figures. Learn the four headline numbers of the 2024 assessment and the safe-versus-over-exploited split. The examiner will change one digit and ask you to spot it.

- Dynamic Ground Water Resource Assessment 2024: annual recharge 446.90 BCM.
- Annual extractable resource 406.19 BCM; actual extraction 245.64 BCM.
- Stage of extraction for the country: 60.47 per cent.
- Of 6,746 assessment units, 4,951 are 'safe' - about 73.4 per cent.
- 751 units are 'over-exploited' (11.1 per cent), 206 'critical' and 711 'semi-critical'.

## Traditional water harvesting - a matching question waiting to happen

### Structure to state

| Structure              | State / region                 |
|------------------------|--------------------------------|
| Khadin and tanka       | Rajasthan (khadin - Jaisalmer) |
| Johad                  | Alwar, Rajasthan               |
| Kuhl                   | Himachal Pradesh               |
| Zabo                   | Nagaland                       |
| Apatani                | Arunachal Pradesh              |
| Ahar-pyne              | Bihar                          |
| Eri tanks              | Tamil Nadu                     |
| Surangam               | Kerala                         |
| Phad                   | Maharashtra                    |
| Bamboo drip irrigation | Meghalaya                      |

**ASKED IN RAS**

Asked in RAS Pre 2024 as a match-the-following - khadin/Rajasthan, zabo/Nagaland, ahar-pyne/Bihar, kuhl/Himachal. Learn all ten pairs above; RPSC rotates which four it puts in the list.

**River interlinking**

- National Perspective Plan framed in 1980; executed by the National Water Development Agency, set up 1982.
- Ken-Betwa is the first river-link project actually taken up.
- It involves Madhya Pradesh and Uttar Pradesh; the storage is the Daudhan dam on the Ken, in Panna.
- Namami Gange is the umbrella Ganga rejuvenation programme - do not confuse it with a link project.

**CURRENT LINK**

Current link, as of September 2026: the foundation stone of the Ken-Betwa Link Project was laid on 25 December 2024 at Khajuraho and construction of the Daudhan dam is under way. The standing controversy is submergence in the core area of the Panna Tiger Reserve. Expect this either as 'India's first river-link project' or paired with Panna.

**REVISION RECAP**

1. Wells and tube wells irrigate over 60 per cent of India's net irrigated area; canals about 25 per cent; tanks only about 3 per cent.
2. UP has the largest net irrigated area; Punjab has the highest irrigated share of its sown area (about 98 per cent); rice takes the largest share of irrigated land.
3. DVC, 1948 - first multipurpose project of independent India, on the TVA model; Tilaiya and Maithon on the Barakar, Konar and Panchet on the Damodar.
4. Order to remember: DVC 1948, Hirakud 1957, Bhakra 1963, Farakka 1975.

5. Tehri is the tallest dam (260.5 m); Bhakra the highest straight gravity dam (226 m); Hirakud among the longest; Nagarjuna Sagar among the largest masonry dams.
6. Bhakra-Sutlej-Bilaspur, reservoir Gobind Sagar. Rihand-Sonbhadra, reservoir Govind Ballabh Pant Sagar. Keep the two reservoirs apart.
7. Farakka, 1975, diverts water into the Bhagirathi-Hooghly for Kolkata port.
8. Nepal appears in two projects - Kosi (Bihar) and Gandak (Bihar plus UP).
9. Chambal valley = MP plus Rajasthan; four structures - Gandhi Sagar, Rana Pratap Sagar, Jawahar Sagar, Kota barrage.
10. Sardar Sarovar on the Narmada benefits Gujarat, MP, Maharashtra and Rajasthan.
11. Indira Gandhi Canal: Harnai barrage, Rajasthan Feeder 204 km, main canal 445 km, tail at Mohangarh in Jaisalmer; India's longest canal.
12. PMKSY - 1 July 2015, 'Har Khet Ko Pani', components AIBP, HKKP and Watershed; Per Drop More Crop moved to RKVY from 2022-23.
13. Atal Bhujal Yojana - 25 December 2019, Jal Shakti, Rs 6,000 crore, half World Bank, seven states, gram panchayat water security plans.
14. Command Area Development Programme began in 1974-75; National Water Policy first framed in 1987.
15. Groundwater 2024: recharge 446.90 BCM, extraction 245.64 BCM, stage 60.47 per cent, 751 over-exploited units.
16. Water harvesting pairs: khadin/tanka Rajasthan, johad Alwar, kuhl Himachal, zabo Nagaland, apatani Arunachal, ahar-pyne Bihar, eri Tamil Nadu, surangam Kerala, phad Maharashtra, bamboo drip Meghalaya.
17. National Perspective Plan 1980, NWDA 1982, Ken-Betwa the first link taken up - Daudhan dam, MP and UP, foundation stone 25 December 2024.